

Operational Amplifier

Video 8

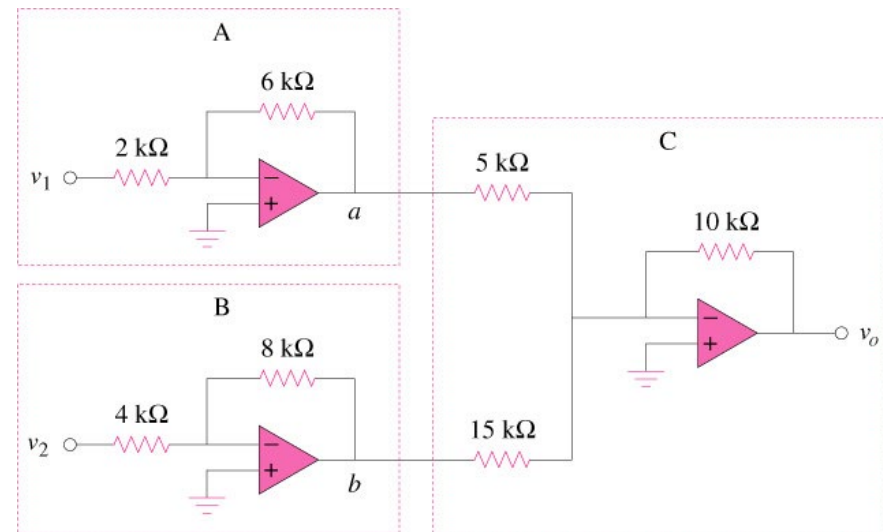
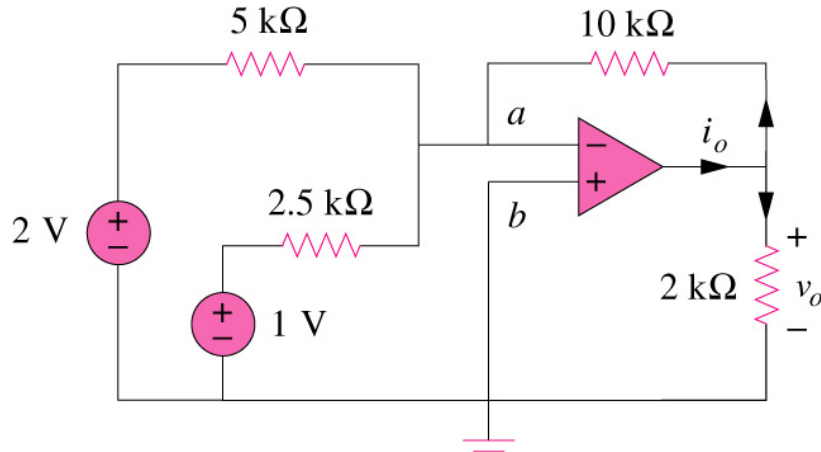
Electrical Science

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University of Central Oklahoma

Operational Amplifier

- Practical and Ideal Op Amp
- Configuration of Op Amp
- Cascaded Op Amp
- Application: Digital-to Analog Converter



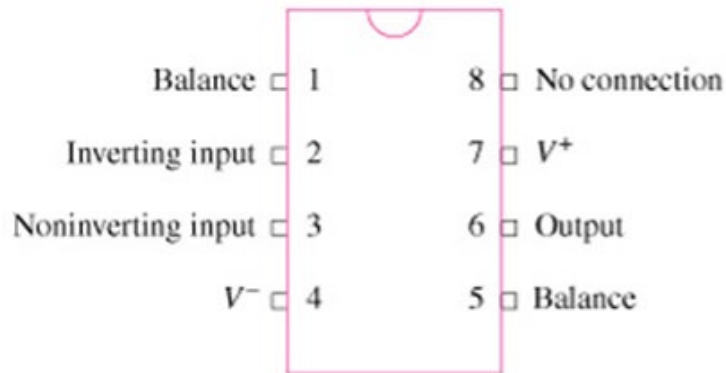
What is an Op Amp?

- It is an electronic unit that consists of transistors, resistors, and capacitors.
- It is an active circuit element designed to perform mathematical operations of ***amplification, addition, subtraction, multiplication, division, differentiation*** and ***integration***.

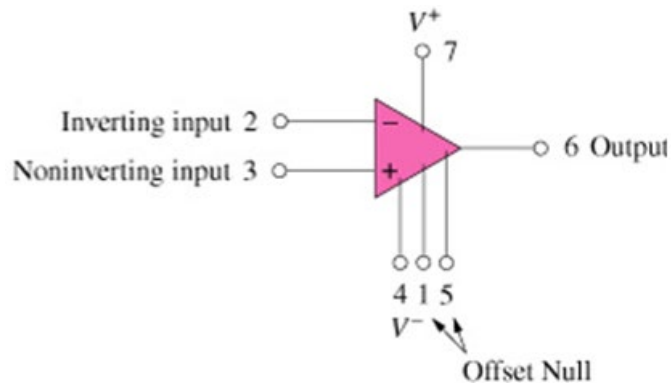
Op Amp Circuit Diagram



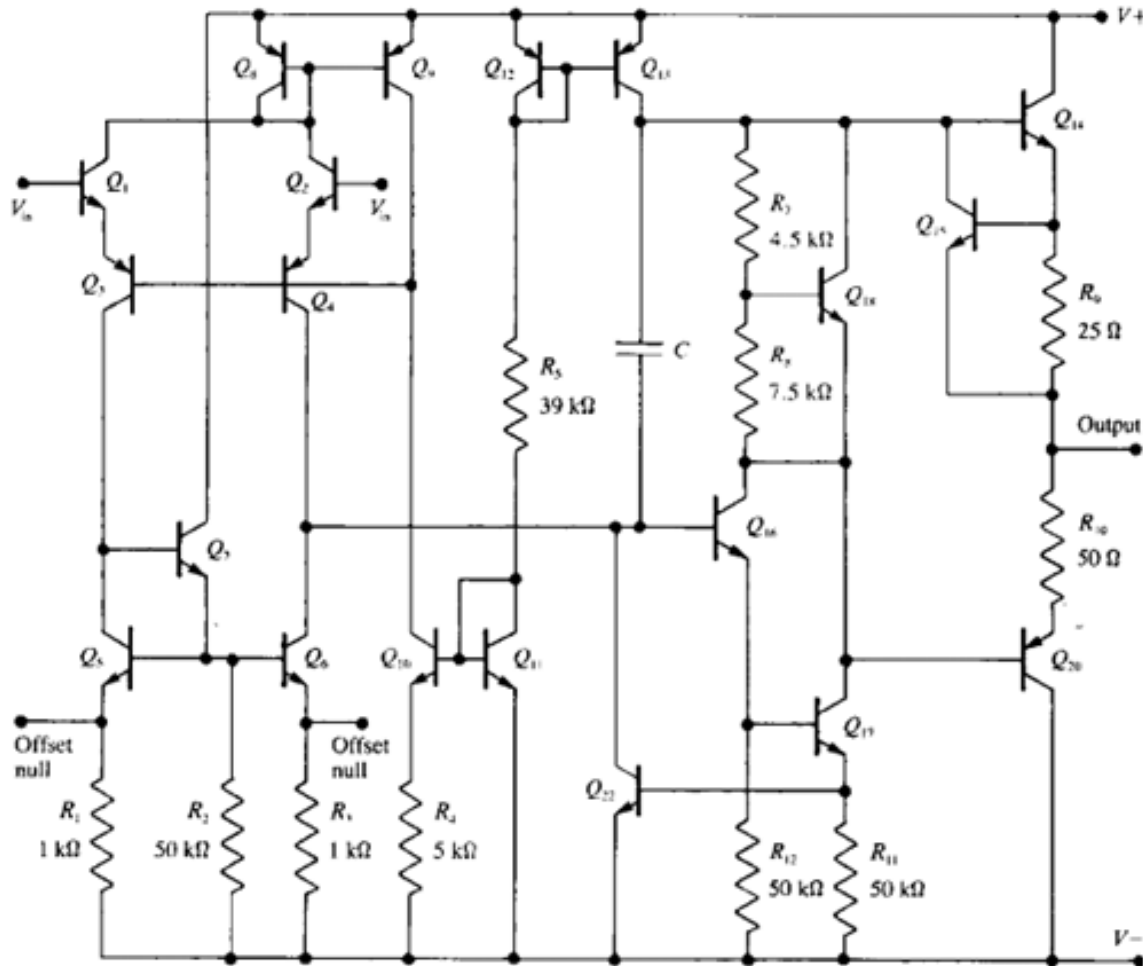
LM 741



Pin Configuration



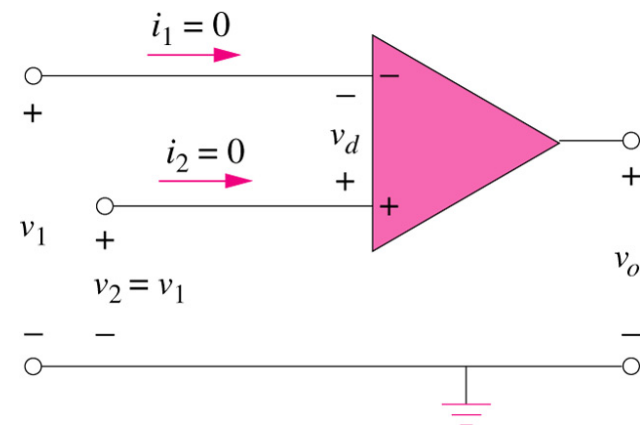
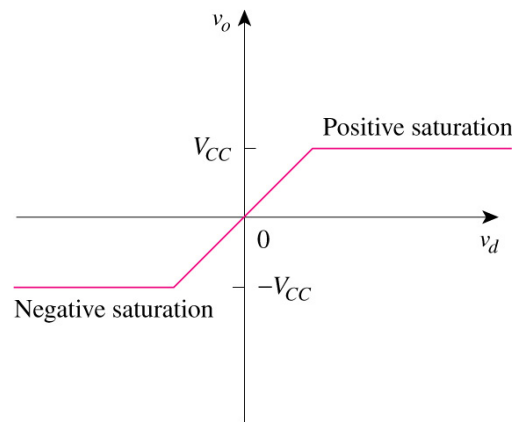
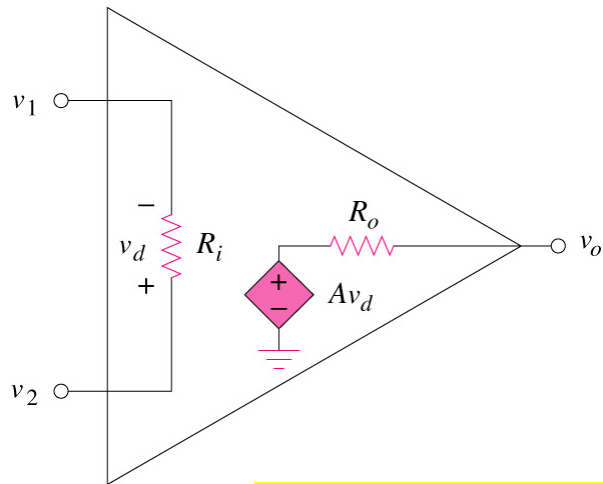
Circuit Symbol



20 Transistors, 11 Resistors, 1 Capacitor

Op Amp Parameters

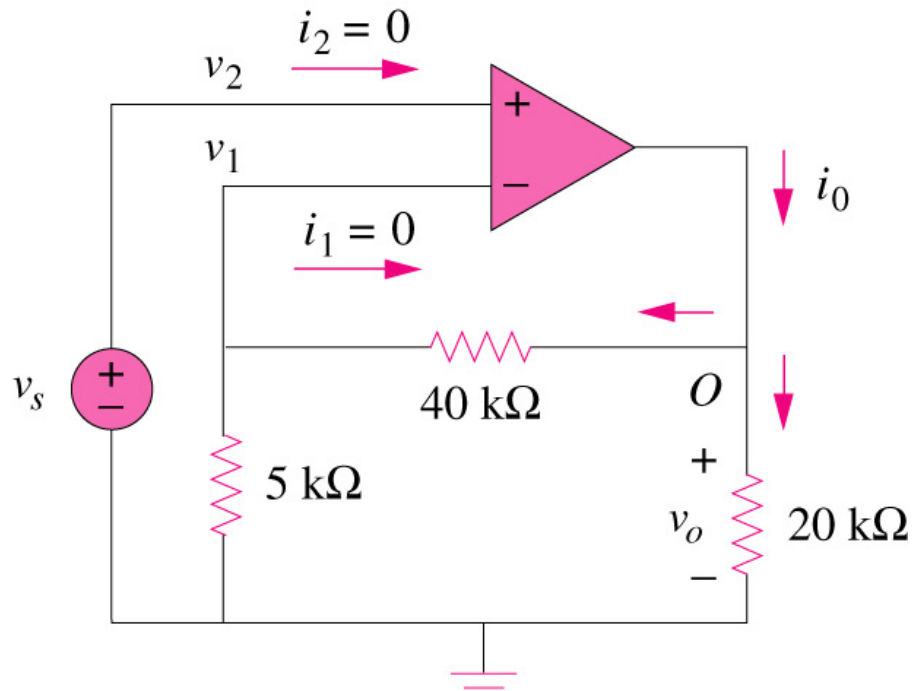
Parameter	Typical range	Ideal values
Open-loop gain, A	10^5 to $10^8 \Omega$	∞
Input resistance, R_i	10^5 to $10^{13} \Omega$	$\infty \Omega$
Output resistance, R_o	10 to 100Ω	0Ω
Supply voltage, V_{CC}	5 to 24 V	



$$v_d = v_2 - v_1; \quad v_o = Av_d = A(v_2 - v_1)$$

Ideal Op Amp Example

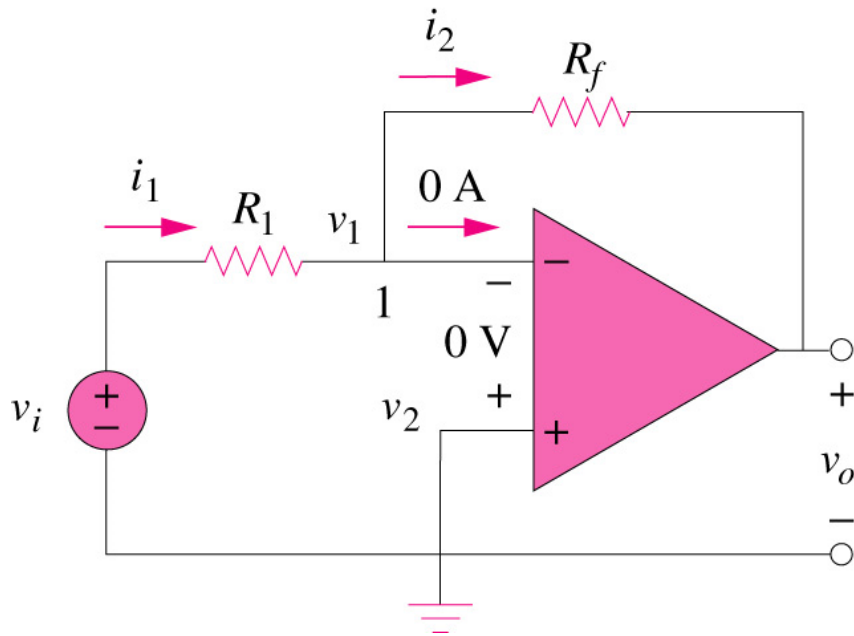
Determine the value of i_o , if $v_s = 1$ v.



Ans: 0.65mA

Inverting Op amp

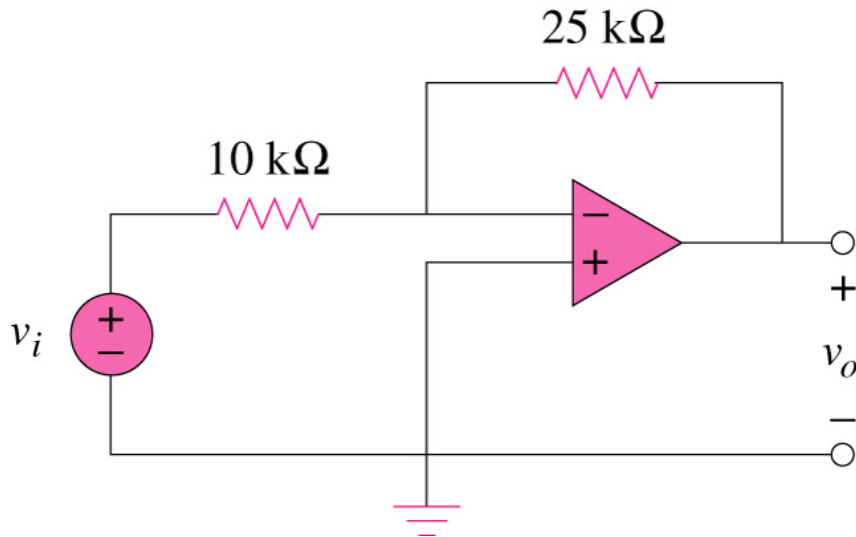
Inverting amplifier reverses the polarity of the input signal while amplifying it



$$v_o = -\frac{R_f}{R_1} v_i$$

Inverting Op amp Example

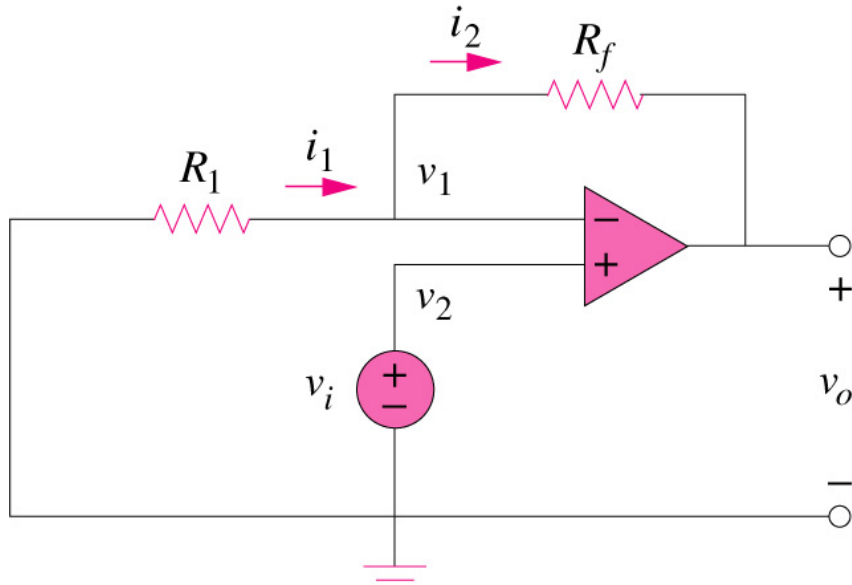
Refer to the op amp below. If $v_i = 0.5\text{V}$, calculate:
(a) the output voltage, v_o and
(b) the current in the $10\text{k}\Omega$ resistor.



Ans: (a) $V_o = -1.25\text{V}$; (b) $50\mu\text{A}$

Non-inverting Op amp

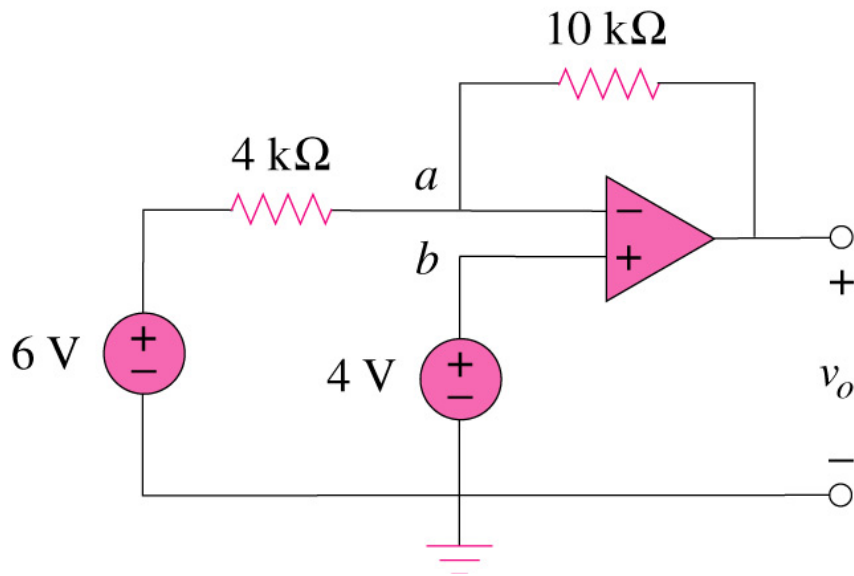
Non-inverting amplifier is designed to produce positive voltage gain



$$v_o = \left(1 + \frac{R_f}{R_1} \right) v_i$$

Configuration of Op amp Example

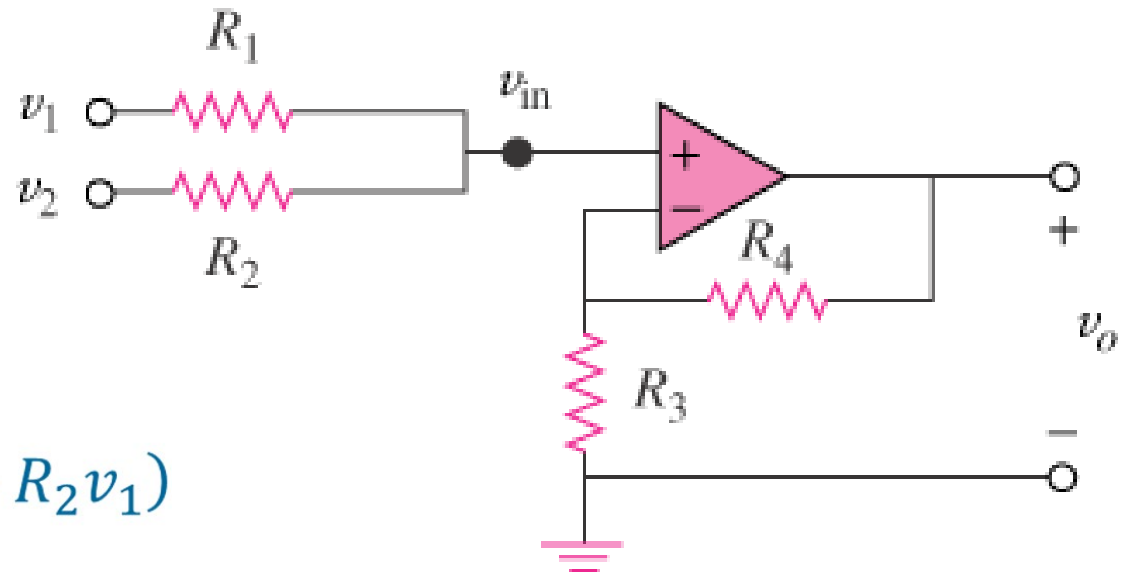
For the op amp shown below, calculate the output voltage v_o .



*Hint: use superposition Ans: -1V

Configuration of Op amp

Given the op amp circuit shown below, express v_o in terms of v_1 and v_2 .



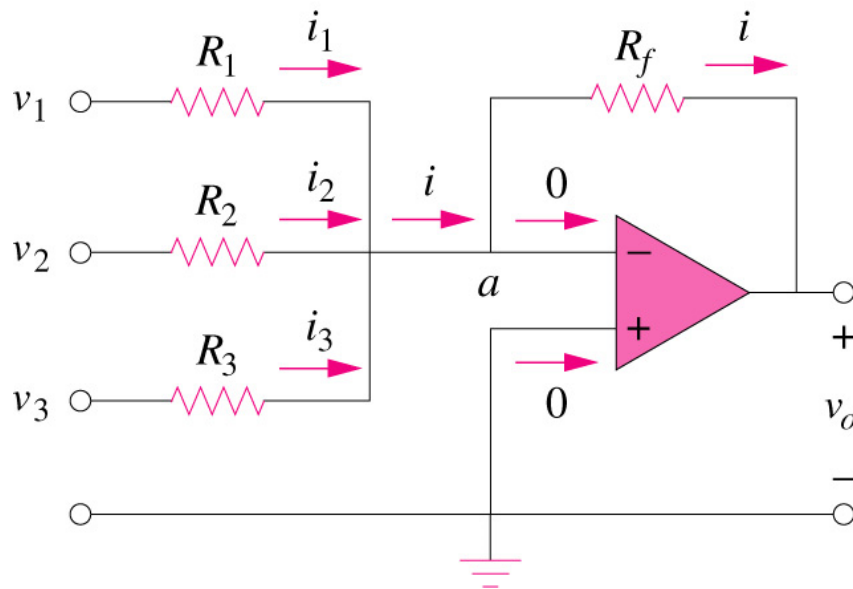
$$v_{in} = \left(\frac{1}{R_1 + R_2} \right) (R_1 v_2 - R_2 v_1)$$

$$v_o = \left(1 + \frac{R_4}{R_3} \right) v_{in}$$

$$v_o = \left(1 + \frac{R_4}{R_3} \right) \left(\frac{1}{R_1 + R_2} \right) (R_1 v_2 - R_2 v_1)$$

Summing Amplifier

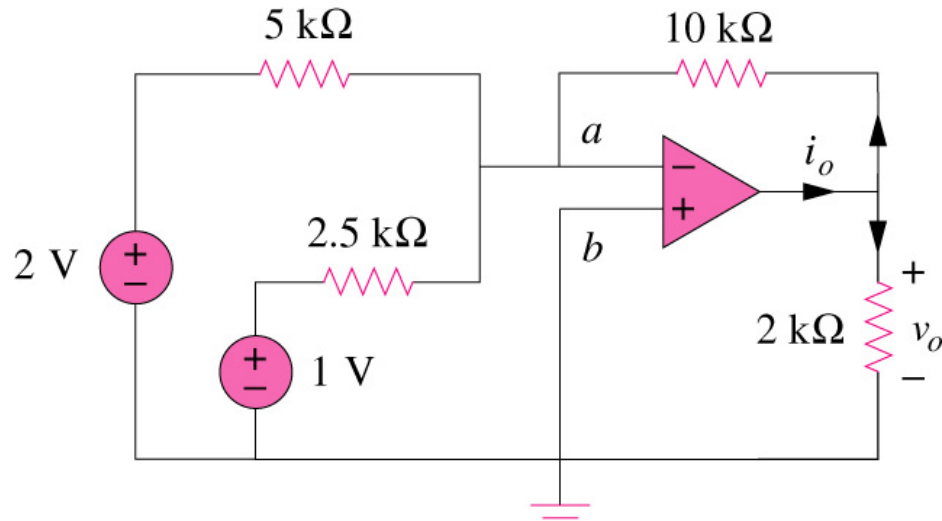
- Summing Amplifier is an op amp circuit that combines several inputs and produces an output that is the weighted sum of the inputs.



$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$

Summing Amplifier Example

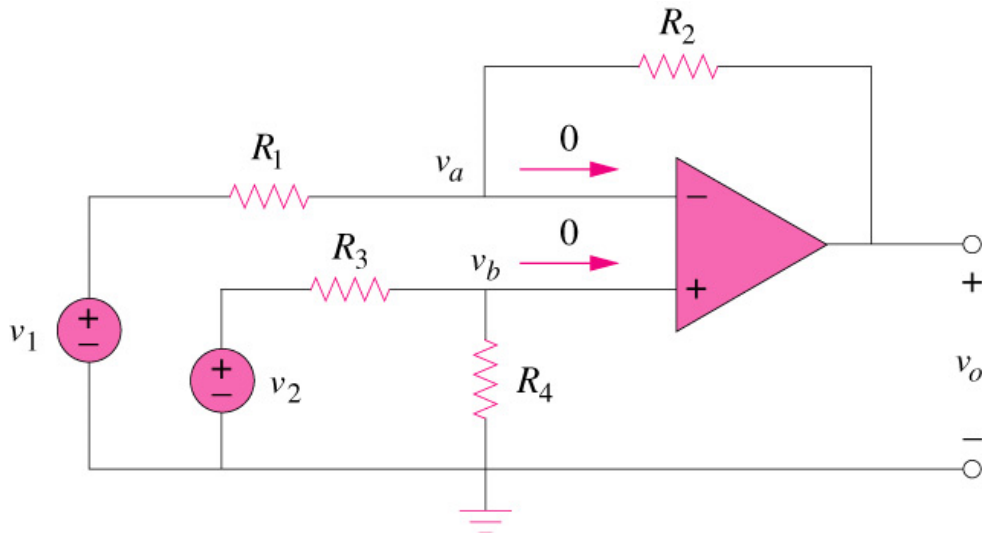
Calculate v_o and i_o in the op amp circuit shown below.



Ans: -8V, -4.8mA

Difference Amplifier

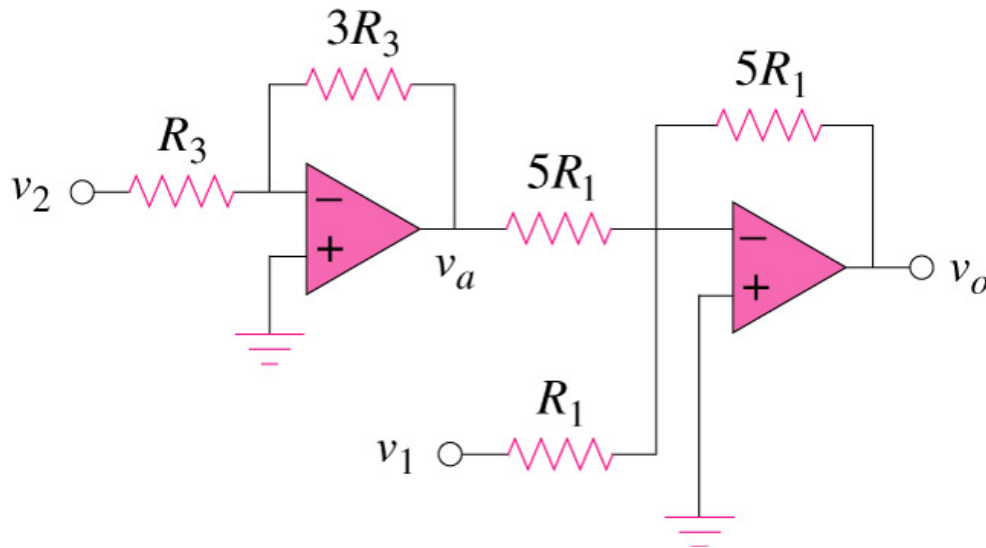
- Difference amplifier is a device that amplifies the difference between two inputs but rejects any signals common to the two inputs.



$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1 \Rightarrow v_o = v_2 - v_1, \text{ if } \frac{R_1}{R_2} = \frac{R_3}{R_4} = 1$$

Difference Op-Amp Example

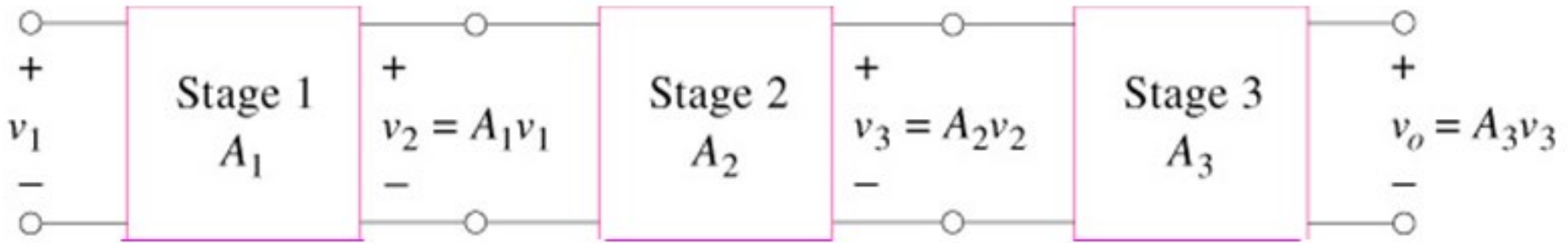
Determine R_1 , R_2 , R_3 and R_4 so that $v_o = -5v_1 + 3v_2$ for the circuit shown below.



Ans: $R_1 = 10\text{k}\Omega$, $R_2 = 50\text{k}\Omega$, $R_3 = 20\text{k}\Omega$, $R_4 = 20\text{k}\Omega$

Cascaded Op Amp

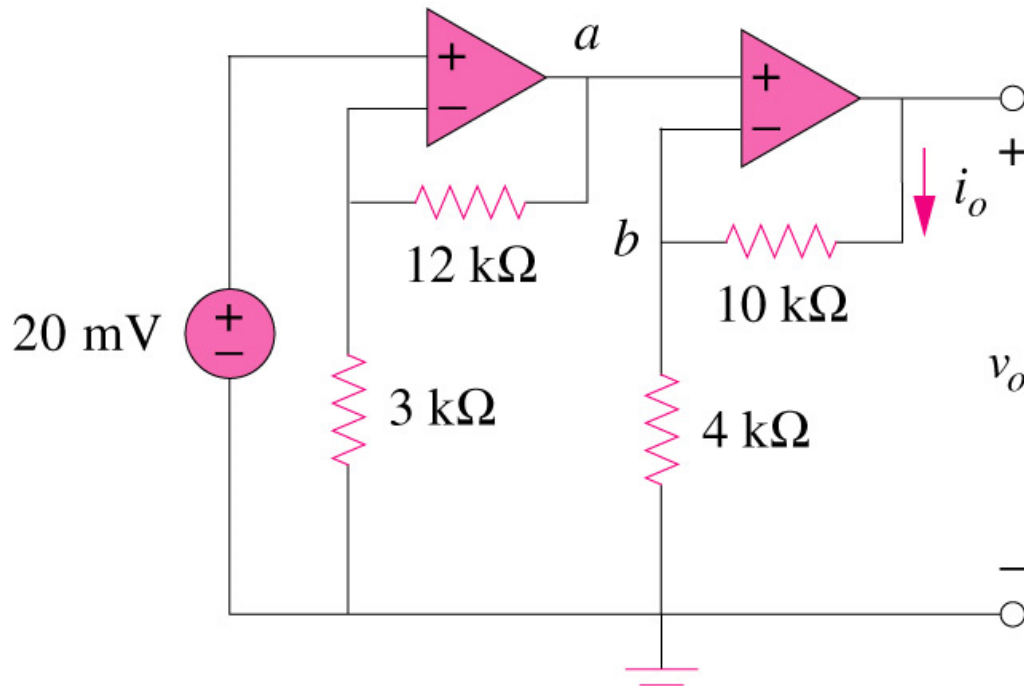
It is a head-to-tail arrangement of two or more op amp circuits such that the output to one is the input of the next.



$$v_o = A_1 A_2 A_3 v_1$$

Cascaded Op Amp Example

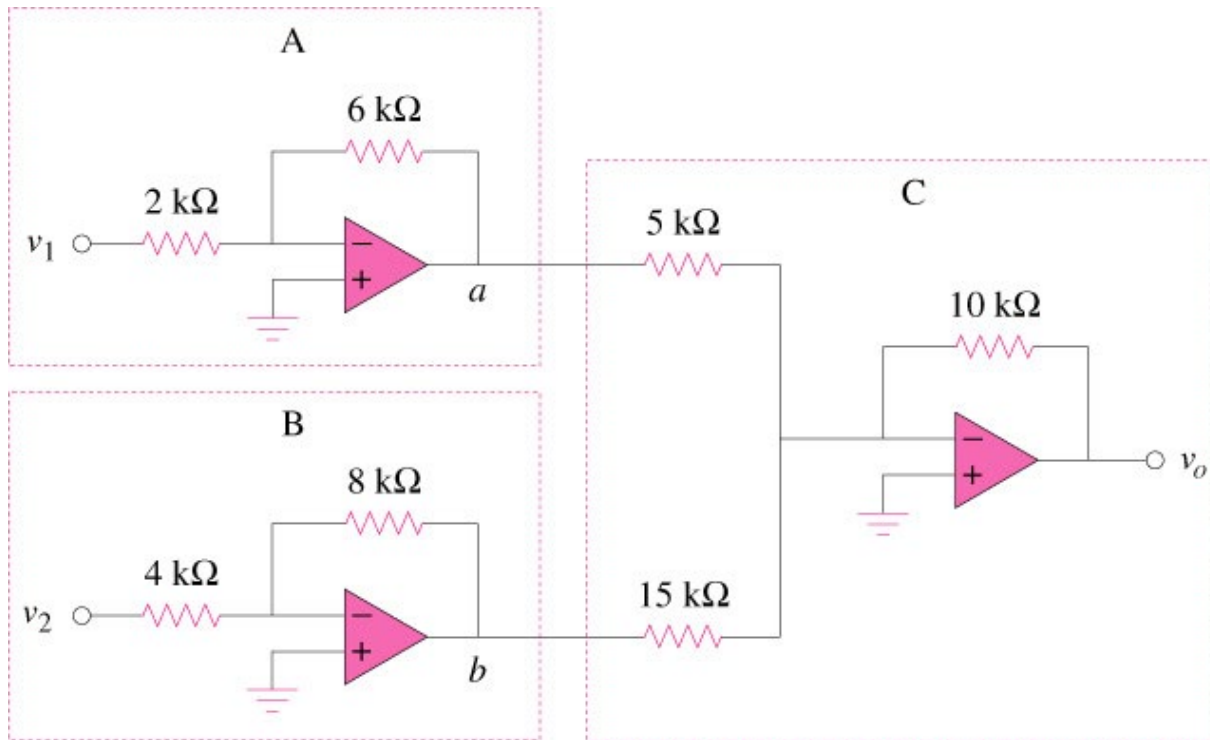
Find v_o and i_o in the circuit shown below.



Ans: 350mV, 25μA

Cascaded and Parallel Op Amp

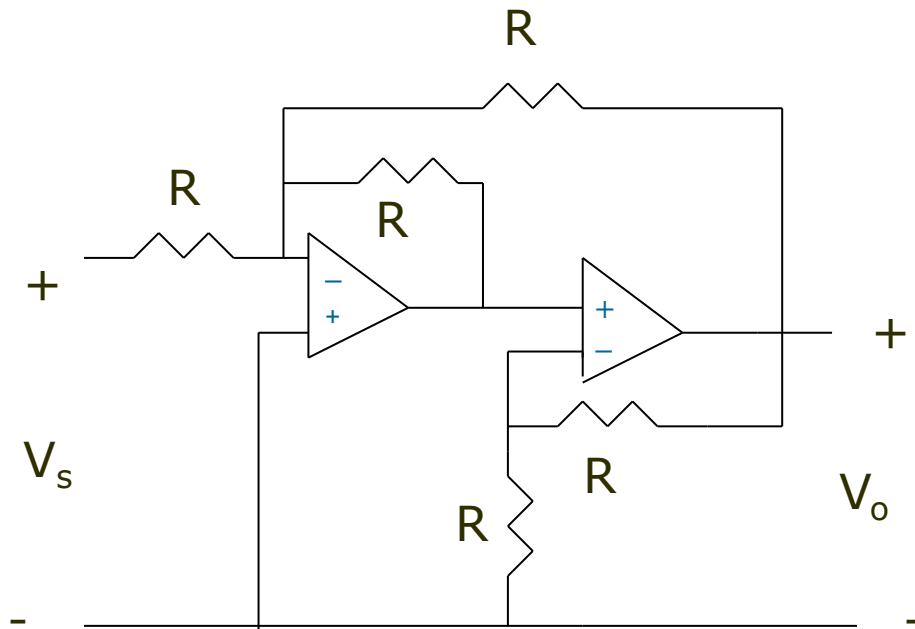
If $v_1 = 1\text{V}$ and $v_2 = 2\text{V}$, find v_o in the op amp circuit shown below.



Ans: 8.667 V

Cascaded Op Amp Example

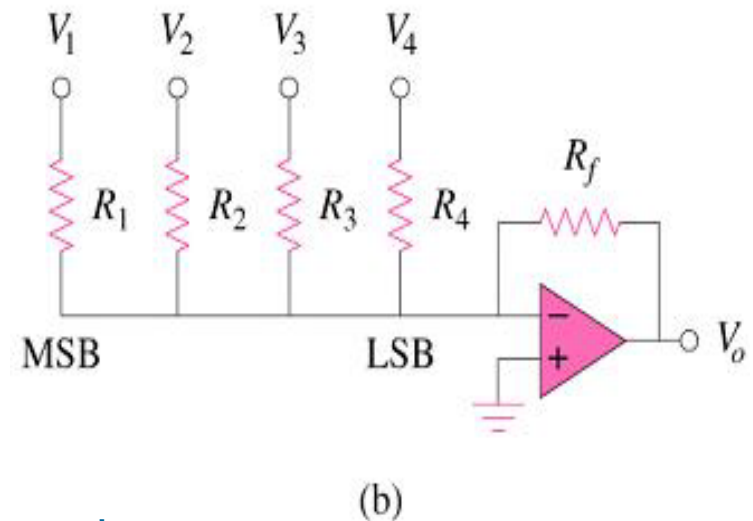
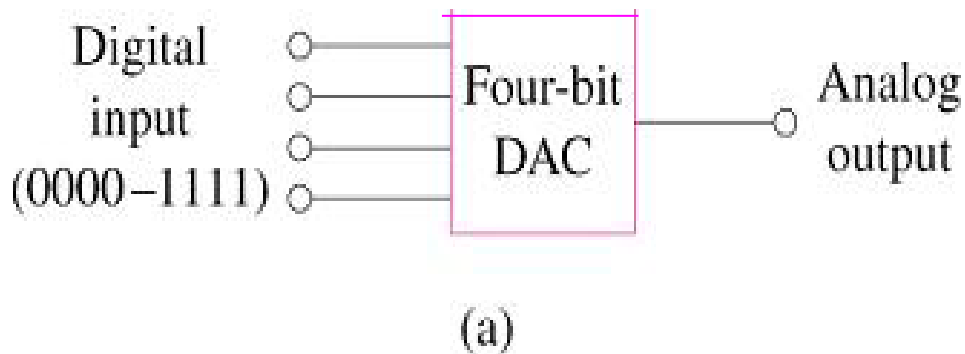
Determine the voltage transfer ratio v_o/v_s in the op amp circuit below, where $R = 10 \text{ k}\Omega$.



Application (DAC)

- Digital-to Analog Converter (DAC) : it is a device which transforms digital signals into analog form.

Four-bit DCA: (a) block diagram (b) binary weighted ladder type



$$-V_o = \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 + \frac{R_f}{R_4} V_4$$

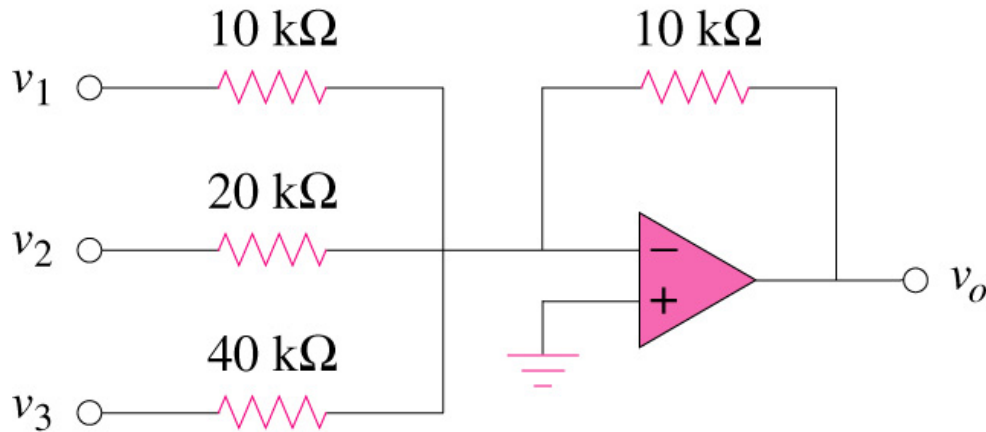
where

V_1 – MSB, V_4 – LSB

V_1 to V_4 are either 0 or 1 V

Application (DAC) Example

For the circuit shown below, calculate v_o if $v_1=0V$, $v_2=1V$ and $v_3 = 1V$, equivalent to 3-bits binary (011).



Ans: $v_o = -0.75V$